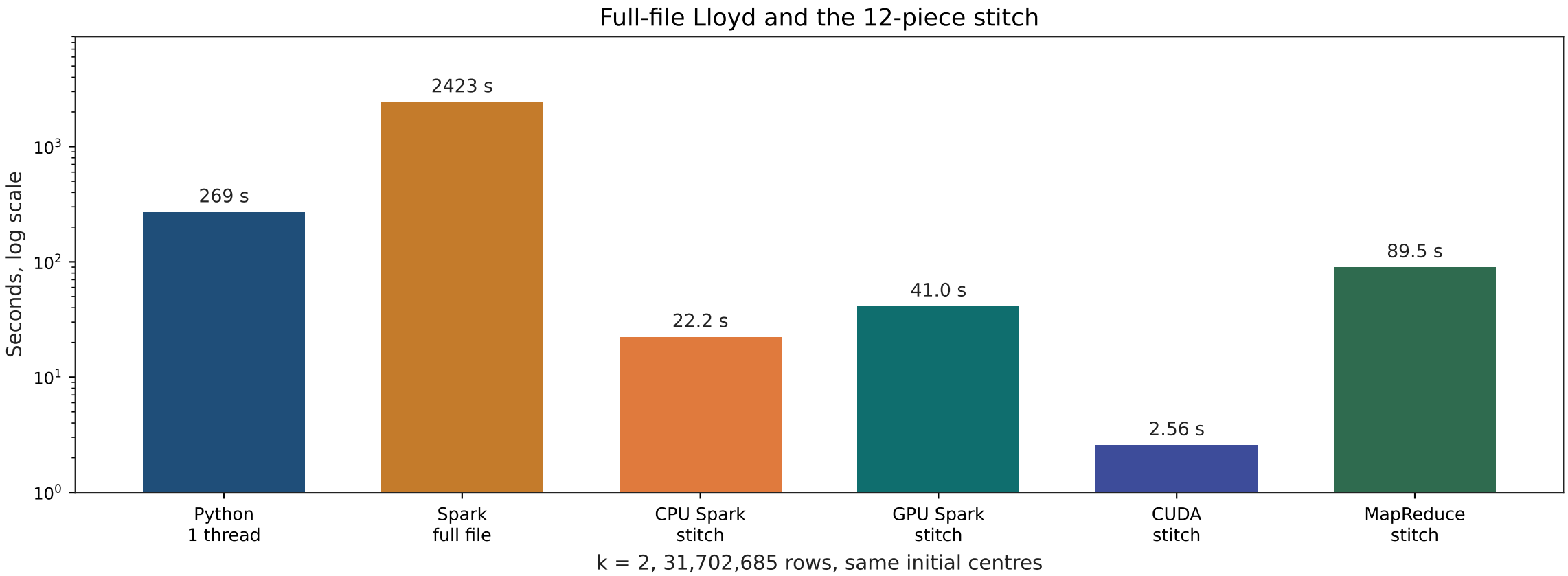

Python is 1 thread, 86 rounds on the whole file. Spark full file is local[12], also 86 rounds.

Stitch: each of 12 pieces clusters alone, then Lloyd on the 24 local centres. Same initial centres, stop at 1e-4.

CPU Spark stitch is local[12], so the twelve pieces run together. Its bar is that one stage, 22.2 s.

GPU Spark stitch is RAPIDS on the RTX 3070, local[1], so the twelve Lloyd loops are added: 41.0 s. Reading each text piece is not in that bar.

CUDA stitch is Lloyd on the same GPU, outside Spark. The copy onto the device, 0.72 s, is not in that bar.

MapReduce is one YARN job, 8 vcores. Putting the pieces on HDFS, 18.0 s, is not in that bar.

The four stitch runs reach the same centres, apart from an eighth-decimal difference on GPU Spark. All four differ from the full-file centres by 1.489.